

Lab 5 Distributed Parallel Computing Assignment

Lab Assignment: Parallel N Queens Problem using OpenMP

Aim

To design and implement a parallel algorithm to solve the **N Queens problem** using OpenMP.

Problem Statement

The N-Queens problem is a classical combinatorial problem in computer science. The objective is to place N queens on an $N \times N$ chessboard such that no two queens attack each other.

Two queens attack each other if they share:

- The same row
- The same column
- The same diagonal

In this assignment, students are required to implement a program using **OpenMP parallel programming** to compute the total number of valid solutions for the **8-Queens problem**.

The program should:

- Place 8 queens on an 8×8 chessboard
- Ensure that no queens attack each other
- Use parallel computation with OpenMP to explore multiple placements
- Calculate the total number of valid solutions
- Measure and display execution time

Example (Understanding the Problem)

For $N = 4$, a valid configuration is:

```
. Q . .  
. . . Q  
Q . . .  
. . Q .
```

Another valid configuration:

```
. . Q .  
Q . . .  
. . . Q  
. Q . .
```

Thus, the total number of solutions for $N = 4$ is **2**.

Expected Output

The program should display:

- Board size ($N = 8$)
- Total number of valid solutions
- Execution time